

# Minimum-variance control of astronomical adaptive optic systems with actuator dynamics under synchronous and asynchronous sampling

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## SUMMARY

Adaptive optic (AO) systems are now routinely used in ground-based telescopes to counter the effects of atmospheric turbulence. A deformable mirror (DM) generates a correction wavefront, which is subtracted from the turbulent wavefront using measurements of the residual phase provided by a wavefront sensor (WFS). Minimizing the variance of the residual phase defines a sampled data control problem combining a continuous time minimum-variance (MV) performance criterion with a discrete-time controller. For a fairly general class of linear time-invariant DM and turbulence WFS models, this control problem can be transformed into an equivalent discrete-time LQ optimization problem involving a set of (discrete-time) control-sufficient statistics of the incoming continuous-time turbulence. This paper shows how to constructively solve this MV problem in the presence of DM's dynamics, starting from continuous-time models of DM and turbulence. This result is extended to the case of asynchronous DM/WFS sampling. An illustrative application to optimal control of tip-tilt turbulent modes for the European extremely large telescope in the presence of first-order DM's dynamics is presented. Copyright © 2010 John Wiley & Sons, Ltd.

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## 1. INTRODUCTION

The main factors limiting the resolution of ground-based telescopes are atmospheric turbulence and wind-induced vibrations, which produce erratic high-frequency deformations in the shape of the optical wavefront entering the telescope's pupil. Because astronomical imaging systems need to integrate the optical flux over long exposure times, these disturbances produce a blurring effect which can be reversed only to a very limited extent through image post-processing. From Galileo's time well into the second half of the 20th century, the only conceivable way to circumvent this problem has been to minimize the amount of turbulent atmosphere itself. This could be achieved to some extent (and at considerable expense) by locating telescopes on top of mountains with favorable weather patterns. Decisive additional improvements can be achieved by inserting a deformable mirror (DM) into the telescope's optical path to counteract in real time, *i.e.* during the very process of image formation, the deformations of the optical wavefront. Although this adaptive optic (AO)

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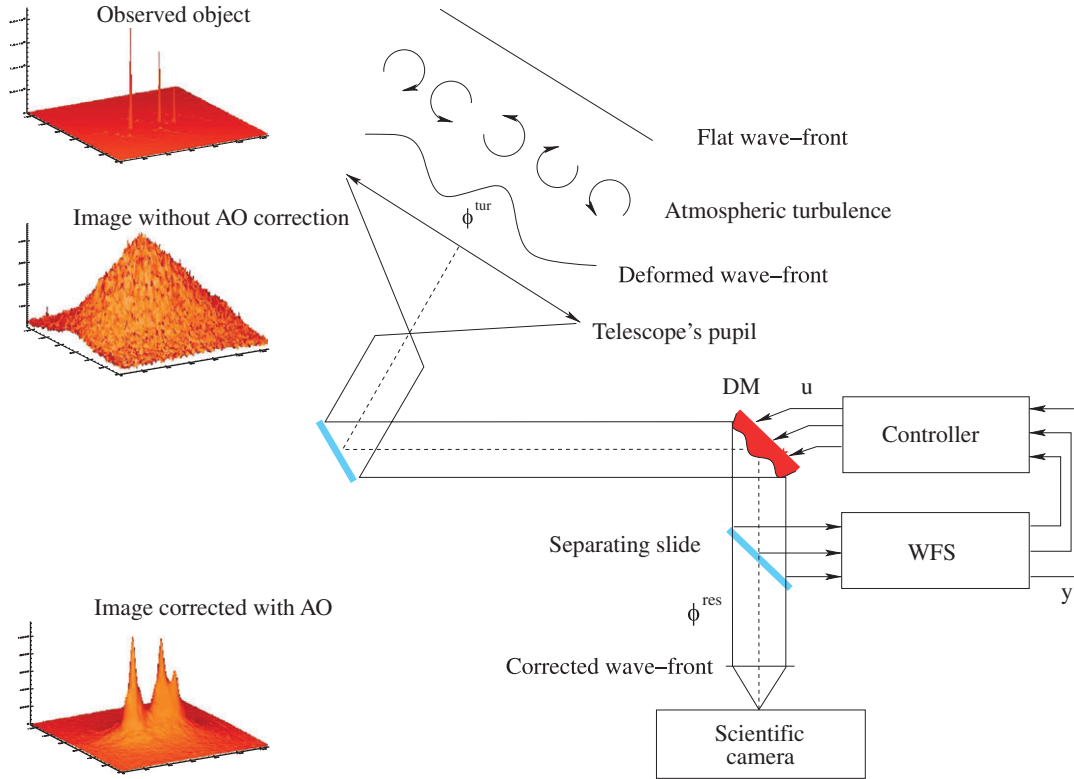


Figure 1. Illustration of the AO principle. Diffraction-limited images are restored by the real-time AO system, in which a wave-front sensor (WFS) measures and a deformable mirror (DM) corrects in real time the distortions affecting wavefronts propagating through the atmosphere. This simple AO configuration features a single DM/WFS pair.

concept can be traced back to the early 1950s [1], the first astronomical AO systems only became operational in the early 1990s [2].

Figure 1 presents the schematics of a standard astronomical AO system. The correction wavefront (the surface deformation  $\phi^{\text{cor}}$  generated by the DM) is subtracted from the turbulent wavefront  $\phi^{\text{tur}}$  entering the telescope's pupil. The DM's control  $u$  is computed by a real-time digital controller and applied through a zero-order hold (ZOH). Discrete measurements of the residual phase  $\phi^{\text{res}} = \phi^{\text{tur}} - \phi^{\text{cor}}$  are provided by a wavefront sensor (WFS). Both devices may operate in either synchronous or asynchronous mode. All wavefront variables are time-varying surfaces, and in practice are represented by finite-dimensional vectors of coordinates in a suitable (truncated) basis of orthogonal functions.

From the control engineering point of view, an AO control obviously corresponds to a disturbance rejection problem. Since, as noted above, astronomical imaging requires an integration over long exposure times, a relevant performance criterion is the averaged variance of the residual phase, defined as

$$J^c(u) \triangleq \lim_{T_f \rightarrow +\infty} \frac{1}{T_f} \int_0^{T_f} \|\phi^{\text{res}}(t)\|^2 dt = \lim_{T_f \rightarrow +\infty} \frac{1}{T_f} \int_0^{T_f} \|\phi^{\text{tur}}(t) - \phi^{\text{cor}}(t)\|^2 dt. \quad (1)$$

Minimizing this continuous-time criterion over a prescribed set of admissible discrete-time control policies  $u = \{u_k\}_{k \in N} = \{u(kT_s)\}_{k \in N}$ , where  $T_s$  is the controller's sampling period, clearly defines a sampled data minimum-variance (MV) optimization problem. (For the sake of simplicity, the notation  $\zeta_k \triangleq \zeta(kT_s)$  will be used throughout this paper for all sampled variables). However, until pretty recently, this sampled data quality of the optimization problem has generally been ignored altogether in the contributions in an MV AO control design. The continuous-time variables in (1)

were often simply replaced by their values at successive sampling times, leading to the simplified discrete-time LQ criterion

$$\begin{aligned} J^{d,s}(u) &\triangleq \lim_{M \rightarrow +\infty} \frac{1}{M} \sum_{k=0}^{M-1} \|\phi^{\text{res}}(kT_s)\|^2 = \lim_{M \rightarrow +\infty} \frac{1}{M} \sum_{k=0}^{M-1} \|\phi_k^{\text{res}}\|^2, \\ &= \lim_{M \rightarrow +\infty} \frac{1}{M} \sum_{k=0}^{M-1} \|\phi_k^{\text{tur}} - \phi_k^{\text{cor}}\|^2. \end{aligned} \quad (2)$$

Starting from pioneering attempts to apply LQG design methods to an AO control [3, 4], many authors have explicitly or implicitly assumed that, for sufficiently small  $T_s$ , the discrete criterion  $J^{d,s}$  (or finite-horizon and/or regularized versions of  $J^{d,s}$ ) constitutes at the very least a sufficiently adequate proxy for  $J^c$ . This approach has been pursued in a number of recent contributions in the field of optimal MV AO control design [5, 6]. However, criteria  $J^c$  and  $J^{d,s}$  are plainly not equivalent, so that minimizing  $J^{d,s}$  in lieu of  $J^c$  inevitably results in a suboptimal controller. Symmetrically, other authors addressed the AO control design problem in a purely continuous-time framework [7]. This also results in suboptimal controllers.

The AO sampled data MV control problem was solved in [8] for a DM with infinitely fast response. It is shown that in order to obtain an equivalent discrete-time LQ criterion the instantaneous values of the phase variables in (2) need to be replaced by their temporal averages over the next sampling interval, as proposed in [9]. This enables to constructively solve the MV optimization problem using the standard LQG/Kalman filtering techniques.

The current generation of DMs used in astronomical applications exhibits effective bandwidths extending well beyond the typical AO loops sampling frequency. For such systems, the ‘infinitely fast DM’ assumption is indeed reasonable. However, the next generation of larger DMs currently under development for the existing Very Large Telescopes (VLTs) and the future Extremely Large Telescopes (ELTs) are likely to exhibit significantly lower resonance frequencies, possibly associated with poorly damped modes. For such systems, controllers designed using a non-optimal discretization of the MV criterion or based on the infinitely fast DM assumption may well result in significant performance degradation. In addition, the performance requirements for future astronomical AO systems, measured in terms of variance reduction, are becoming ever more stringent. Investigating the sampled data MV problem for DMs with dynamics has therefore acquired more pressing urgency.

A constructive solution to the sampled-data MV problem can be found in [10, 11], under the assumptions that continuous-time LTI state-space models are available for both the turbulent phase and for the DM, and that the optimal controller is also an LTI (discrete-time) system. The solution relies on the construction of an equivalent discrete-time LQ criterion for an augmented system, using the standard lifting techniques from sampled data theory. Yet this approach does not fully exploit the particular structure of this sampled data disturbance rejection problem, and especially the fact that the plant to be controlled is the combination of a stochastic disturbance model and of a deterministic actuator model.

An alternative approach is to derive an equivalent discrete-time MV criterion by directly assessing the effect of control decisions on the value of  $J^c$ . In [12, 13], it was shown that a completely equivalent and well-posed MV criterion can be constructed in this way for a wide class of continuous-time DM LTI dynamics, and independently of any particular assumption on the turbulent phase. The equivalent criterion centrally involves, in addition to the DM’s internal state at sampling instants, two other discrete variables: the average of  $\phi^{\text{tur}}$  and another, weighted average of  $\phi^{\text{tur}}$  over the next sampling interval. This weighted average can be interpreted as a dual variable encapsulating the infra-sampling period interaction between the turbulence and the DM’s dynamics. The MV control problem can then be solved constructively, using the standard LQG techniques, as long as a stochastic LTI model for these averages is available.

For AO systems, the discrete-time signals derived from  $\phi^{\text{tur}}$  appearing in the equivalent discrete criterion define a sufficient statistic, as defined, e.g. in [14], in both the full information and the incomplete information cases. A self-contained proof is presented for the procedure used in [12, 13]

to derive a control sufficient stochastic discrete-time model from a stochastic continuous-time model of  $\phi^{\text{tur}}$ . This construction is then used to constructively solve the MV control problem in the case of asynchronous WFS/DM sampling, i.e. when the total time  $T_c$  allocated for measurement processing and control computation is not an integral multiple of  $T_s$ .

A solution to the asynchronous problem has been proposed in [10, 11] based on the approach and assumptions described above. The asynchronous problem had been addressed hitherto only in the infinitely fast DM case, and subject to additional restrictive assumptions. Thus, an optimal solution was obtained in [8] and [15] on the condition that  $T_c$  and  $T_s$  are commensurable, while simpler suboptimal solutions were proposed for arbitrary values of  $T_c$ ,  $T_s$  in [16] and [17]. The approach presented here allows to derive a fully optimal solution of the general asynchronous sampled data MV problem for arbitrary values of  $T_c$ ,  $T_s$  and any choice of DM's dynamics.

This paper is organized as follows. Sections 2 and 3 recall the construction of the equivalent criterion, sufficient statistic and of optimal controllers for the simpler infinitely fast DM case and for a general class of LTI DM's dynamics. Section 4 shows how to construct, based on continuous-time DM and turbulence models, a stochastic discrete-time model enabling to compute the optimal discrete-time controller in reconstructed feedback form, and also to evaluate both the optimal value of  $J^c$  and the performance of suboptimal controllers using the standard LQG techniques. Section 5 extends these results to asynchronous WFS/DM sampling. In Section 6, an illustrative application to the control of a tip-tilt mirror with first-order dynamics is presented, in conditions representative of the future European-ELT. Some perspectives for future research are outlined in Section 7.

## 2. MV CONTROL FOR INFINITELY FAST DM

In the 'infinitely fast DM' case, the DM's settling time is assumed to be negligible with respect to the sampling period  $T_s$ . When the control  $u_k = u(kT_s)$  is applied through a ZOH, then for  $kT_s \leq t < (k+1)T_s$ , the correction phase is  $\phi^{\text{cor}}(t) = Nu_k$ . The DC gain matrix  $N$  is the so-called DM influence matrix, each column defining the influence function of (the surface deformation generated by) a particular actuator. Without loss of generality (since this property can always be recovered by altering the definition of the control input), it shall be assumed in the sequel that  $N$  is full column-rank, i.e. that  $N^T N$  is invertible.

Breaking the integral in (1) into a sum of successive temporal slices with length  $T_s$ , we get

$$J^c(u) = \lim_{M \rightarrow +\infty} \frac{1}{M} \sum_{k=0}^{M-1} J_k^c(u), \quad (3)$$

where the incremental cost is

$$J_k^c(u) \triangleq \frac{1}{T_s} \int_{kT_s}^{(k+1)T_s} \|\phi^{\text{res}}(t)\|^2 dt. \quad (4)$$

For an infinitely fast DM, this becomes

$$J_k^c(u) = \frac{1}{T_s} \int_{kT_s}^{(k+1)T_s} \|\phi^{\text{tur}}(t) - Nu_k\|^2 dt. \quad (5)$$

Let  $\bar{\phi}_{k+1}^{\text{tur}}$  be the average of  $\phi^{\text{tur}}$  over the sampling period  $[kT_s, (k+1)T_s]$ :

$$\bar{\phi}_{k+1}^{\text{tur}} \triangleq \frac{1}{T_s} \int_{kT_s}^{(k+1)T_s} \phi^{\text{tur}}(t) dt. \quad (6)$$

Because  $\phi^{\text{tur}} - \bar{\phi}_{k+1}^{\text{tur}}$  is by construction a zero-mean variable over  $[kT_s, (k+1)T_s]$ ,

$$J_k^c(u) = \frac{1}{T_s} \int_{kT_s}^{(k+1)T_s} \|\phi^{\text{tur}}(t) - \bar{\phi}_{k+1}^{\text{tur}}\|^2 dt + \|\bar{\phi}_{k+1}^{\text{tur}} - Nu_k\|^2 \triangleq \delta J_k^c + J_k^d(u), \quad (7)$$

where

$$J_k^d(u) \triangleq \|\bar{\phi}_{k+1}^{\text{tur}} - Nu_k\|^2. \quad (8)$$

Consider now the discrete-time infinite-horizon criterion

$$J^d(u) \triangleq \lim_{M \rightarrow +\infty} \frac{1}{M} \sum_{k=0}^{M-1} J_k^d(u) = \lim_{M \rightarrow +\infty} \frac{1}{M} \sum_{k=0}^{M-1} \|\bar{\phi}_{k+1}^{\text{tur}} - Nu_k\|^2. \quad (9)$$

As  $\delta J_k^c$  is not affected by  $u$ , for any set  $\mathcal{U}$  of admissible control policies,  $\arg \min_{\mathcal{U}} J^d = \arg \min_{\mathcal{U}} J^c$ , so that  $J^d$  is an equivalent discrete-time LQ criterion for the original sampled data MV problem. Furthermore, because  $u_k$  does not affects  $J_{k'}^d$  for  $k' \neq k$ , the optimal control can be computed as a simple orthogonal projection. Under the full information hypothesis that the value of  $\bar{\phi}_{k+1}^{\text{tur}}$  is known at time  $k$ , the optimal control would be the least-squares solution

$$u_k^* = (N^T N)^{-1} N^T \bar{\phi}_{k+1}^{\text{tur}}. \quad (10)$$

This establishes that  $\bar{\phi}_{k+1}^{\text{tur}}$  is indeed a sufficient statistics for the full information MV control problem.

In practice, however,  $u_k$  should be computed from delayed and noisy measurements available at time  $t = kT_s$ . Because we are now dealing with a standard discrete-time LQ problem, the separation property is valid. In other words, assuming that both  $\bar{\phi}_{k+1}^{\text{tur}}$  and the measurements are second-order random variables, the optimal MV incomplete information control becomes

$$u_k^* = (N^T N)^{-1} N^T \widehat{\bar{\phi}_{k+1}^{\text{tur}}}, \quad (11)$$

where  $\widehat{\bar{\phi}_{k+1}^{\text{tur}}}$  denotes the conditional expectation of  $\bar{\phi}_{k+1}^{\text{tur}}$  with respect to the sequence

$$\mathcal{Y}_k \triangleq \{y_0, y_1, \dots, y_k\} \quad (12)$$

of all measurements available at time  $t = kT_s$ . When the dynamics of  $\bar{\phi}_{k+1}^{\text{tur}}$  and its relation to WFS measurements can be modeled using a standard LTI discrete-time state-space model driven by Gaussian white noises, the solution of this MV optimal prediction problem is given by a Kalman filter [8, 9].

At this point, it is worth mentioning that the construction of this model is facilitated by the fact that wavefront sensors work by integrating the residual phase over successive sampling intervals (or portions thereof). As a result, in a typical AO configuration where WFS and DM operations are synchronized, and where one sampling period is allocated for the measurement processing and control computations, a linear WFS measurement naturally takes the form

$$y_k = y(kT_s) = D \left( \frac{1}{T_s} \int_{(k-2)T_s}^{(k-1)T_s} \phi^{\text{res}}(t) dt \right) + w_k \quad (13)$$

$$= D \bar{\phi}_{k-1}^{\text{tur}} - DNu_{k-2} + w_k, \quad (14)$$

where  $D$  is a device-specific phase-to-measurements linear operator.

As a consequence, a discrete-time stochastic model for  $\bar{\phi}^{\text{tur}}$  is all that is needed to constructively solve the MV control problem in *both* the full information and incomplete information cases. Such a model could be derived from an underlying continuous-time model of  $\phi^{\text{tur}}$ , as shown in [18]. A simpler vector-valued AR(1) model of the form

$$\bar{\phi}_{k+1}^{\text{tur}} = A^{\text{tur}} \bar{\phi}_k^{\text{tur}} + v_k \quad (15)$$

has been proposed in [9]. LQG controllers based on this model that enable to reproduce with reasonable accuracy the short-term spatio-temporal correlation structure of the sufficient statistic  $\bar{\phi}^{\text{tur}}$ , have been successfully tested on realistic AO simulators and on the ONERA test benches BOA and HOMER [19, 20].

### 3. EQUIVALENT LQ CRITERION AND SUFFICIENT STATISTICS FOR DMS WITH DYNAMICS

This construction of a discrete-time LQ criterion equivalent to  $J^c$  can be extended to LTI DM's dynamics described by a state-space model in the general form

$$\dot{x}_m(t) = A_m x_m(t) + B_m u(t), \quad (16)$$

$$p(t) = C_m x_m(t) + D_m u(t), \quad (17)$$

$$\phi^{\text{cor}}(t) = N p(t) \quad (18)$$

and under the following assumptions:

- (i)  $A_m$  is a stability matrix
- (ii) The pair  $(A_m, C_m)$  is observable
- (iii) The DC gain between  $u$  and  $p$  is unitary, i.e.  $D_m - C_m A_m^{-1} B_m = I$
- (iv) The matrix  $N$  is full column rank—or equivalently,  $N^T N$  is invertible.

This formalism and assumptions actually define a fairly general class of DM's dynamics. One important special case is when the DM's response can be represented by a finite element model (FEM) description in the form

$$\phi^{\text{cor}}(t) = \Psi_\phi^T \eta(t), \quad (19)$$

where  $\eta(t)$  is the solution of the matrix-valued second-order equation

$$M_\eta \ddot{\eta}(t) + K_\eta \dot{\eta}(t) + D_\eta \eta(t) = \Psi_e e(t). \quad (20)$$

Here,  $e(t)$  denotes the DM's input signal, while  $M_\eta$  is an invertible matrix (for the control of DM surface using FEM models, see, e.g. [7]). These equations can be rewritten in state-space form as

$$\begin{pmatrix} \dot{\eta}(t) \\ \ddot{\eta}(t) \end{pmatrix} = \begin{pmatrix} 0 & I \\ -M_\eta^{-1} K_\eta & -M_\eta^{-1} D_\eta \end{pmatrix} \begin{pmatrix} \eta(t) \\ \dot{\eta}(t) \end{pmatrix} + \begin{pmatrix} 0 \\ M_\eta^{-1} \Psi_e \end{pmatrix} e(t), \quad (21)$$

$$\phi^{\text{cor}}(t) = \Psi_\phi^T \eta(t). \quad (22)$$

Let us now make the following assumptions: (a) This FEM model is internally stable; (b) the matrix  $\Psi_e$  has full row rank, i.e.  $\text{rank}(\Psi_e) = \dim(\eta)$ —or equivalently,  $\Psi_e \Psi_e^T$  is invertible; (c) the matrix  $\Psi_\phi$  has full row rank, i.e.  $\text{rank}(\Psi_\phi) = \dim(x)$ —or equivalently,  $\Psi_\phi \Psi_\phi^T$  is invertible.

Consider now the system with output  $\phi^{\text{cor}}$ , a new input  $u$  with dimension  $\dim(u) = \dim(\eta)$ , and intermediate output  $p$  defined through the identities

$$e(t) \triangleq \Psi_e^T u(t), \quad x_m(t) \triangleq \begin{pmatrix} \eta(t) \\ \dot{\eta}(t) \end{pmatrix}, \quad p(t) \triangleq (\Psi_e \Psi_e^T)^{-1} K_\eta \eta(t). \quad (23)$$

Note that assumption (a) necessarily implies that  $K_\eta$  is invertible (otherwise, the state transition matrix would have null eigenvalues). Thus, this new system admits a state-space representation in the form (16)–(18), with

$$A_m = \begin{pmatrix} 0 & I \\ -M_\eta^{-1} K_\eta & -M_\eta^{-1} D_\eta \end{pmatrix}, \quad B_m = \begin{pmatrix} 0 \\ M_\eta^{-1} \Psi_e \Psi_e^T \end{pmatrix}, \quad (24)$$

$$C_m = ((\Psi_e \Psi_e^T)^{-1} K_\eta \quad 0), \quad D_m = 0, \quad N = \Psi_\phi^T K_\eta^{-1} \Psi_e \Psi_e^T. \quad (25)$$

Using routine matrix calculations, it is readily checked that assumptions (i)–(iv) are verified. Also note that this construction could be extended to a 'mixed' case where only a limited number of

eigenmodes are retained in the FEM description, while the remaining ones are assumed to have instantaneous response [13].

We are now in a position to state the main result of this section:

*Proposition 1*

Let the AO control be applied through a ZOH with sampling period  $T_s$  to a DM with dynamics in the form (16)–(18) verifying assumptions (i)–(iv). Then, for any set  $\mathcal{U}$  of admissible discrete-time control strategies and for all  $\lambda \geq 0$ ,  $\arg \min_{\mathcal{U}} J^c = \arg \min_{\mathcal{U}} J^d$ , where

$$J^d(u) \triangleq \lim_{M \rightarrow +\infty} \frac{1}{M} \sum_{k=0}^{M-1} (z_k^T Q_1 z_k + 2z_k^T S_1 u_k + u_k^T R_1 u_k) \quad (26)$$

with

$$z_k \triangleq \begin{pmatrix} \overline{\phi}_{k+1}^{\text{tur}} \\ \overline{\phi}_{k+1}^{\text{tur}} \\ x_{m,k} \end{pmatrix} \triangleq \begin{pmatrix} \frac{1}{T_s} \int_{kT_s}^{(k+1)T_s} \phi^{\text{tur}}(t) dt \\ \frac{1}{T_s} \int_{kT_s}^{(k+1)T_s} e^{(t-kT_s)A_m^T} C_m^T N^T \phi^{\text{tur}}(t) dt \\ x_{m,k}(kT_s) \end{pmatrix} \quad (27)$$

$$Q_1 \triangleq \begin{pmatrix} 2\lambda N N^T & 0 & 0 \\ 0 & \lambda (I + 2B_m^T A_m^{-T} A_m^{-1} B_m) & -I \\ 0 & -I & Q_m \end{pmatrix}, \quad S_1 \triangleq \begin{pmatrix} -N \\ -A_m^{-1} B_m \\ S_m \end{pmatrix}, \quad (28)$$

$$R_1 \triangleq R_m \triangleq \frac{1}{T_s} \int_0^{T_s} (I + C_m e^{tA_m} A_m^{-1} B_m)^T N^T N (I + C_m e^{tA_m} A_m^{-1} B_m) dt, \quad (29)$$

$$S_m \triangleq \frac{1}{T_s} \int_0^{T_s} e^{tA_m^T} C_m^T N^T N (I + C_m e^{tA_m} A_m^{-1} B_m) dt, \quad (30)$$

$$Q_m \triangleq \frac{1}{T_s} \int_0^{T_s} e^{tA_m^T} C_m^T N^T N C_m e^{tA_m} dt. \quad (31)$$

In addition:

- (1) For  $\lambda = 0$ ,  $J^c(u) = J^d(u) + \lim_{T_f \rightarrow +\infty} \frac{1}{T_f} \int_0^{T_f} \|\phi^{\text{tur}}(t)\|^2 dt$ .
- (2) For large enough values of  $\lambda$ ,  $Q_1 - S_1 R_1^{-1} S_1^T \geq 0$ , so that  $J^d(u) \geq 0$  for all  $u$ .

*Proof*

See [12]. □

*Remarks*

- (1) When  $z_k$  can be obtained as the output of a standard stochastic LTI model, the fact that  $Q_1 - S_1 R_1^{-1} S_1^T \geq 0$  for large enough values of the arbitrary constant  $\lambda$  guarantees that the sampled data MV problem has a unique solution, which can be expressed in standard reconstructed feedback form. However, it is easily checked that as  $\lambda$  increases, the largest eigenvalue of  $Q_1$  inevitably tends to  $+\infty$ , which may lead to numerical instabilities in solving the associated ARE. For this reason, it might be preferable instead to feed into the LQ solver the non-positive value of  $Q_1$  corresponding to  $\lambda = 0$ , provided of course that the said solver tolerates it (as does the MATLAB routine `dlqr`).
- (2) More generally, the sole purpose of assumptions (i)–(iv) is ultimately to ensure the existence of a unique solution to the discrete LQ control problem. If some or all of them are relaxed, the construction can be modified to guarantee a unique solution for a regularized version of

the original MV criterion in the form

$$J^c(u) \triangleq \lim_{T_f \rightarrow +\infty} \frac{1}{T_f} \int_0^{T_f} (\|\phi^{\text{res}}(t)\|^2 + u^T(t) R_u u(t)) dt \quad (32)$$

with  $R_u > 0$ .

- (3) As indicated above, proposition 1 can be viewed as a natural extension of the design procedure in Section 2 for the infinitely fast case. Conversely, the infinitely fast DM case is recovered as a limit case of Proposition 1. To prove it, consider a set of DM models in the form (16)–(18) depending on a parameter  $\theta$  such that for  $\theta \rightarrow 0$ , the maximum value of the real part of all eigenvalues of  $A_m$  tend to  $-\infty$ , so that the dynamics, loosely speaking, become ‘infinitely fast’. The corresponding asymptotic behavior for the weighting matrices in Proposition 1 is then

$$\lim_{\theta \rightarrow 0} R_m = N^T N, \quad \lim_{\theta \rightarrow 0} S_m = 0, \quad \lim_{\theta \rightarrow 0} Q_m = 0. \quad (33)$$

Likewise, as  $\theta \rightarrow 0$ , the weighting function used to define the statistics  $\bar{\phi}^{\text{tur}}$  also uniformly converges to zero over the interval  $[0, T_s]$ . Assuming that  $\phi^{\text{tur}}$  is a stationary process with finite variance,  $\bar{\phi}^{\text{tur}}$  then also converges toward zero in a stochastic  $\mathcal{L}_2$  sense, with both its mean and variance converging toward zero. Thus, in the limit case  $\theta = 0$ , one may remove both  $x_{m,k}$  and  $\bar{\phi}_{k+1}^{\text{tur}}$  from  $J_k^d$ . Taking  $\lambda = 1$ , this results in the same discrete criterion as in Section 2.

- (4) From a control-theoretical point of view,  $\bar{\phi}^{\text{tur}}$  and  $\bar{\phi}^{\text{tur}}$  can be interpreted as dual control variables, since they formally derive from manipulations of the scalar product  $\langle x, y \rangle \triangleq \frac{1}{T_s} \int_0^{T_s} x(t)^T y(t) dt$ . The formulae defining  $\bar{\phi}^{\text{tur}}$  and  $\bar{\phi}^{\text{tur}}$  thus correspond to convolution integrals in retrograde time. In order to express them as the output of a causal system, as will be done in the following section, it will therefore be necessary, so to speak, to reverse time once more and to include in the process appropriate anti-causal (and unstable) operators.

A direct corollary of Proposition 1 is that the pair  $(\bar{\phi}^{\text{tur}}, \bar{\phi}^{\text{tur}})$  is a sufficient statistic for the sampled data AO MV control problem in the full information case. It also constitutes a sufficient statistic in the incomplete information case. Assuming, as in Section 2, that WFS and DM operations are synchronized and a measurement processing and control delay of one sampling period, one gets from (13)

$$y_k = D \bar{\phi}_{k-1}^{\text{tur}} - DN \frac{1}{T_s} \int_{(k-2)T_s}^{(k-1)T_s} p(t) dt + w_k. \quad (34)$$

The control  $u$  being applied through a ZOH, the surface deformation trajectory between successive sampling times is given for all  $t = kT_s + s$  by

$$p(kT_s + s) = C_m e^{sA_m} x_{m,k} + \left( \int_0^s C_m e^{tA_m} B_m ds + D_m \right) u_k. \quad (35)$$

Using the facts that  $A_m$  is invertible and that  $D_m - C_m A_m^{-1} B_m = I$ , this becomes

$$p(t) = p(kT_s + s) = C_m e^{sA_m} x_{m,k} + (I + C_m e^{sA_m} A_m^{-1} B_m) u_k. \quad (36)$$

Substituting this expression in (34) and evaluating the resulting integrals, one gets

$$\begin{aligned} y_k = & D \bar{\phi}_{k-1}^{\text{tur}} - \frac{1}{T_s} DN C_m (e^{T_s A_m} - I) A_m^{-1} x_{m,k-2} \\ & - DN \left( I + \frac{1}{T_s} C_m (e^{T_s A_m} - I) A_m^{-2} B_m \right) u_{k-2} + w_k. \end{aligned} \quad (37)$$

In other words,  $y$  can be expressed as a linear combination of properly delayed versions of  $u$ , of the DM's internal state  $x_m$ , and of  $\bar{\phi}^{\text{tur}}$ , plus the measurement noise.

## 4. DISCRETE MODEL FOR THE CONTROL-SUFFICIENT STATISTICS

When a state-space LTI model can be constructed for the sufficient statistics  $(\overline{\phi}^{\text{tur}}, \overline{\varphi}^{\text{tur}})$ , it is clear from Proposition 1 that the MV controller can be obtained as the solution of a standard discrete-time LQ optimal control problem. As in the infinitely fast DM case, such a model can be either directly inferred from a combination of physical considerations or from the information extracted from empirical measurements.

Alternatively, in this section, we show how such a model can be derived from an underlying stable continuous-time stochastic turbulence model in the standard general form

$$dx_{\phi}(t) = A_{\phi}x_{\phi}(t) dt + d\eta(t), \quad (38)$$

$$\phi^{\text{tur}}(t) = C_{\phi}x_{\phi}(t), \quad (39)$$

where  $\eta$  is a Gaussian white noise with power spectral density (PSD)  $\Sigma_{\eta}$ , independent from the measurement noise  $w$ . Without loss of generality, since for the purpose of infinite-horizon MV control what only matters is the existence of asymptotically stationary solutions, we shall assume that  $x_{\phi}(0)=0$ .

*Proposition 2*

For the DM model (16)–(18), the measurement model (34) and the turbulence model (38)–(39), and assuming that  $A_{\phi}$  is a stability matrix, there exists a Gaussian white noise  $v$  independent of  $w$ , with variance  $\text{Var}(v_k) \triangleq \Sigma_v$ , and matrices  $A, B, \Gamma, G, C$  such that

$$x_{k+1} = Ax_k + Bu_k + \Gamma v_k, \quad (40)$$

$$z_k = Gx_k, \quad (41)$$

$$y_k = Cx_{k-1} + w_k, \quad (42)$$

where

$$x_k \triangleq \begin{pmatrix} x_{\phi,k+1} \\ \overline{\phi}_{k+1}^{\text{tur}} \\ \overline{\varphi}_{k+1}^{\text{tur}} \\ \overline{\phi}_k^{\text{tur}} \\ x_{m,k} \\ x_{m,k-1} \\ u_{k-1} \end{pmatrix}. \quad (43)$$

Expressions for  $A, B, \Gamma, G, C$  and  $\Sigma_v$  are given in Appendix A.

*Proof*

The values of  $G$  and  $C$  flow directly from the definition of  $z$  and from the measurement equation (34). The parts of  $A, B$  describing the discrete dynamics of  $x_m$  result from the standard discretization formulae. Once discrete delays have been accounted for, the only remaining non-trivial part of the construction is the derivation of the equivalent discrete model for the sufficient statistics  $\overline{\phi}^{\text{tur}}$  and  $\overline{\varphi}^{\text{tur}}$ , including the covariance matrix  $\Sigma_v$ .

This is achieved by applying to the turbulence model (38)–(39) which is essentially an extension of a realization procedure initially developed by Åström for modeling averaging sensors [21] and applied more recently by Looze to CCD-based sensors [18]. Consider first a stochastic state-space

transition equation in the form

$$\begin{aligned} d\zeta(t) &\triangleq \begin{pmatrix} d\zeta_1(t) \\ d\zeta_2(t) \end{pmatrix} = \begin{pmatrix} A_\phi & 0 \\ \frac{1}{T_s} e^{-T_s \Theta} \Lambda C_\phi & \Theta \end{pmatrix} \begin{pmatrix} \zeta_1(t) \\ \zeta_2(t) \end{pmatrix} dt + \begin{pmatrix} I \\ 0 \end{pmatrix} d\eta(t) \\ &\triangleq A_\zeta \zeta(t) dt + \Gamma_\zeta d\eta(t) \end{aligned} \quad (44)$$

with the initial condition  $\zeta(0)=0$  and where  $\Lambda$  and  $\Theta$  are matrices whose values shall be specified below. Because the state transition equation for  $\zeta_1$  is the carbon copy of the stochastic model (38), with the same zero initial condition, its solution is  $\zeta_1(t)=x_\phi(t)$ . The stochastic process  $\zeta_2$  can be expressed as a convolution integral of the turbulent phase:

$$\zeta_2(t) = \frac{1}{T_s} \int_0^t e^{(t-s)\Theta} e^{-T_s \Theta} \Lambda C_\phi x_\phi(s) ds = \frac{1}{T_s} \int_0^t e^{(t-s-T_s)\Theta} \Lambda \phi^{\text{tur}}(s) ds. \quad (45)$$

At sampling times  $t=kT_s$ , this becomes

$$\zeta_{2,k} = \frac{1}{T_s} e^{(k-1)T_s \Theta} \int_0^{kT_s} e^{-t\Theta} \Lambda \phi^{\text{tur}}(t) dt. \quad (46)$$

Discretizing model (44) for the sampling period  $T_s$  therefore yields the discrete-time stochastic model

$$\zeta_{k+1} = e^{T_s A_\zeta} \zeta_k + v'_k, \quad (47)$$

where  $v'$  is a discrete Gaussian white noise with variance

$$\text{Var}(v'_k) \triangleq \Sigma_{v'} = \int_0^{T_s} e^{tA_\zeta} \begin{pmatrix} \Sigma_\eta & 0 \\ 0 & 0 \end{pmatrix} e^{tA_\zeta^T} dt. \quad (48)$$

As  $A_\zeta$  is bloc-triangular, so is  $e^{tA_\zeta}$ , and the discrete-time transition equation (47) can be broken down as

$$\zeta_{1,k+1} = e^{T_s A_\phi} \zeta_{1,k} + v'_{1,k}, \quad (49)$$

$$\zeta_{2,k+1} = \Omega \zeta_{1,k} + e^{T_s \Theta} \zeta_{2,k} + v'_{2,k}, \quad (50)$$

where  $\Omega$  is the bottom-left corner of  $e^{T_s A_\zeta}$ , i.e.

$$e^{T_s A_\zeta} = \begin{pmatrix} e^{T_s A_\phi} & 0 \\ \Omega & e^{T_s \Theta} \end{pmatrix}. \quad (51)$$

Considering now the change of variables

$$\zeta'_{1,k} \triangleq \zeta_{1,k+1} = x_{\phi,k+1}, \quad (52)$$

$$\zeta'_{2,k} \triangleq \zeta_{2,k+1} - e^{T_s \Theta} \zeta_{2,k}. \quad (53)$$

The dynamics of these new variables is

$$\zeta'_{1,k+1} = e^{T_s A_\phi} \zeta'_{1,k} + v_{1,k}, \quad (54)$$

$$\zeta'_{2,k+1} = \Omega \zeta'_{1,k} + v_{2,k} \quad (55)$$

with  $v_k = v'_{k+1}$ . On the other hand, from (46) and the definition of  $\zeta'_{2,k}$  one gets

$$\zeta'_{2,k} = \frac{1}{T_s} e^{kT_s \Theta} \int_{kT_s}^{(k+1)T_s} e^{-t\Theta} \Lambda \phi^{\text{tur}}(t) dt = \frac{1}{T_s} \int_{kT_s}^{(k+1)T_s} e^{(kT_s-t)\Theta} \Lambda \phi^{\text{tur}}(t) dt. \quad (56)$$

Finally, take

$$\Theta \triangleq \begin{pmatrix} 0 & 0 \\ 0 & -A_m^T \end{pmatrix}, \quad \Lambda \triangleq \begin{pmatrix} I \\ C_m^T N^T \end{pmatrix}. \quad (57)$$

With this particular choice of  $\Theta$ , it is immediately checked that

$$e^{t\Theta} \Lambda = \begin{pmatrix} I \\ e^{-tA_m^T} C_m^T N^T \end{pmatrix} \quad (58)$$

so that

$$\zeta'_{2,k} = \begin{pmatrix} \frac{1}{T_s} \int_{kT_s}^{(k+1)T_s} \phi^{\text{tur}}(t) dt \\ \frac{1}{T_s} \int_{kT_s}^{(k+1)T_s} e^{(t-kT_s)A_m^T} C_m^T N^T \phi^{\text{tur}}(t) dt \end{pmatrix} = \begin{pmatrix} \bar{\phi}_{k+1}^{\text{tur}} \\ \bar{\varphi}_{k+1}^{\text{tur}} \end{pmatrix}. \quad (59)$$

This discrete model thus provides a stochastic realization for the sufficient statistics  $\bar{\phi}^{\text{tur}}$ ,  $\bar{\varphi}^{\text{tur}}$ , leading directly to formulae in Appendix A for the upper part of  $A$ ,  $B$ .  $\square$

Using this model, the MV controller can be obtained in a standard reconstructed feedback form, *i.e.* by applying the optimal full information control gain to the optimal estimate of  $x_k$  given by a Kalman filter. Because the measurement equation (42) includes a delay of one sampling period, it is immediately checked that in this case, the update equation for the MV estimate  $\hat{x}_{k|k}$  corresponds to the standard predictor form of the Kalman filter. In other words, the equations for the MV controller correspond to

$$u_k = -K_\infty \hat{x}_{k|k}, \quad (60)$$

$$\hat{x}_{k|k} = A \hat{x}_{k-1|k-1} + B u_{k-1} + L_\infty (y_k - C \hat{x}_{k-1|k-1}). \quad (61)$$

The optimal stationary controller and observer gains  $K_\infty$  and  $L_\infty$  are computed from the solutions of appropriate control and filtering AREs. Since the criterion  $J^d$  is defined as a function of  $u$  and  $z = Gx$ , the optimal control gain is

$$K_\infty \triangleq (R_1 + B^T P_\infty B)^{-1} (B^T P_\infty A + G S_1), \quad (62)$$

where  $P_\infty$  is the unique solution of ARE

$$P_\infty = G^T Q_1 G + A^T P_\infty A - (A^T P_\infty B + S_1^T G^T) (R_1 + B^T P_\infty B)^{-1} (B^T P_\infty A + G S_1). \quad (63)$$

Likewise, the optimal observer gain is given by

$$L_\infty \triangleq A \Sigma_\infty C^T (\Sigma_w + C \Sigma_\infty C^T)^{-1}, \quad (64)$$

where the asymptotic estimation error covariance matrix  $\Sigma_\infty$  is the unique solution of ARE

$$\Sigma_\infty = \Gamma \Sigma_v \Gamma^T + A \Sigma_\infty A^T - A \Sigma_\infty C^T (\Sigma_w + C \Sigma_\infty C^T)^{-1} C \Sigma_\infty A^T. \quad (65)$$

(Note that due to the special structure of this disturbance-rejection MV problem, the system  $(A, B, C)$  is structurally neither controllable nor observable. Both the control and estimation AREs nevertheless have a unique solution since  $A$  is a stability matrix).

The discrete model also enables to evaluate directly the optimal value of  $J^c$ , using the following standard result from LQG theory: for any stationary stable system in the form

$$x_{k+1} = A_f x_k + \Gamma_f \mu_k, \quad (66)$$

where  $\mu$  is a white noise sequence with variance  $\Sigma_\mu$ , and for any matrix  $Q_f \geq 0$ , the almost sure (a.s.) identity holds:

$$\lim_{M \rightarrow +\infty} \frac{1}{M} \sum_{k=0}^{M-1} x_k^T Q_f x_k \stackrel{\text{a.s.}}{=} \text{trace}(Q_f \Sigma_x), \quad (67)$$

where  $\Sigma_x$  is the covariance matrix for  $x_k$ , i.e. the unique solution of the Lyapunov equation

$$\Sigma_x = A_f \Sigma_x A_f^T + \Gamma_f \Sigma_\mu \Gamma_f^T. \quad (68)$$

To apply these formulae, it suffices to recall that from Proposition 1

$$J^c(u) = \lim_{T_f \rightarrow +\infty} \frac{1}{T_f} \int_0^{T_f} \|\phi^{\text{tur}}(t)\|^2 dt + \lim_{M \rightarrow +\infty} \frac{1}{M} \sum_{k=0}^{M-1} (z_k^T Q_0 z_k + 2z_k^T S_1 u_k + u_k^T R_1 u_k), \quad (69)$$

where  $Q_0$  is the value of  $Q_1$  corresponding to  $\lambda=0$ :

$$Q_0 \triangleq \begin{pmatrix} 0 & 0 & 0 \\ 0 & 0 & -I \\ 0 & -I & Q_m \end{pmatrix}. \quad (70)$$

As  $\phi^{\text{tur}}$  is the solution of (38)–(39), it is a stationary ergodic zero-mean process, so that

$$\lim_{T_f \rightarrow +\infty} \frac{1}{T_f} \int_0^{T_f} \|\phi^{\text{tur}}(t)\|^2 dt \stackrel{\text{a.s.}}{=} \text{Var}(\phi^{\text{tur}}(t)) \stackrel{\text{a.s.}}{=} \lim_{M \rightarrow +\infty} \frac{1}{M} \sum_{k=0}^{M-1} \|\phi_k^{\text{tur}}\|^2. \quad (71)$$

As a consequence, the value of this term can also be recovered from the steady-state closed-loop covariance of  $x$ . The rest of the construction is straightforward. Take

$$x_k \triangleq \begin{pmatrix} x_k \\ \hat{x}_{k|k} \end{pmatrix}, \quad \mu_k \triangleq \begin{pmatrix} v_k \\ w_k \end{pmatrix}. \quad (72)$$

Using this notation, the closed-loop dynamics for the controller (60)–(61) corresponds to (66), with

$$A_f \triangleq \begin{pmatrix} A & -BK_\infty \\ L_\infty C & A - BK_\infty - L_\infty C \end{pmatrix}, \quad \Gamma_f \triangleq \begin{pmatrix} \Gamma \\ L_\infty \end{pmatrix}. \quad (73)$$

This leads to

$$J^c(u) \stackrel{\text{a.s.}}{=} \text{trace}(Q_f \Sigma_x), \quad (74)$$

where  $\Sigma_x$  is the solution of (68) and

$$Q_f \triangleq \begin{pmatrix} G^T Q_0 G + Q_\phi & S_1 K_\infty \\ K_\infty^T S_1^T & K_\infty^T R_1 K_\infty \end{pmatrix} \quad (75)$$

with

$$Q_\phi \triangleq (C_\phi 0 \ 0 \ 0 \ 0 \ 0 \ 0)(C_\phi 0 \ 0 \ 0 \ 0 \ 0 \ 0)^T. \quad (76)$$

Note that this approach can be used also when  $K_\infty$  and/or  $L_\infty$  are replaced by non-optimal controller and observer gains, and also when erroneous values  $A'$ ,  $B'$ ,  $C'$  are used to compute the observer. In this case,  $A_f$  should be replaced by

$$A'_f \triangleq \begin{pmatrix} A & -B'K \\ LC & A' - B'K - LC' \end{pmatrix}. \quad (77)$$

This enables to evaluate also the degradation in the performance for a wide class of suboptimal controllers, and especially the infinitely fast DM controller in Section 2. The capacity to evaluate expected controller performance is of course important at the design stage. It should also be noted that the value of  $J^c$  can be recovered by measuring what astronomers call the Strehl ratio, which is derived from a long-exposure image of a point-like light source (e.g. a star). In the absence of turbulence, this image would form a distinctive pattern, the Airy spot. In the presence of turbulence, the blurred image of a point source is larger.

The Strehl ratio  $S_r$ , which is defined as the ratio between central peak values of point spread functions of the blurred image and the Airy spot, is thus a decreasing function of  $J^c$ , with  $S_r = 1$  for  $J^c = 0$ ; and for small enough values of  $J^c$ , the approximation  $S_r \simeq \exp(-J^c)$  holds. This allows to assess with good accuracy the value of  $J^c$  actually achieved by an operational astronomical AO system.

## 5. ASYNCHRONOUS DM/WFS SAMPLING

In this section, it shall be assumed that while the DM and WFS still operate with the same sampling period  $T_s$ , they are no longer synchronized, with controls applied at times  $t = kT_s$  and measurements taken at times  $t = kT_s - T_c$ . Without loss of generality, it shall be assumed that  $0 < T_c < T_s$ , since the more general case can be dealt with by incorporating extra discrete delays in the model. In practice,  $T_c$  corresponds to the total time allocated for the measurement processing and control computation.

In the discrete-time control formulation used so far in this paper,  $y_k$  has denoted the last measurement available for computing the control  $u_k = u(kT_s)$ . With this convention, the model for the asynchronous case corresponds to the modified measurement equation

$$y_k = D \left( \frac{1}{T_s} \int_{(k-1)T_s - T_c}^{kT_s - T_c} \phi^{\text{res}}(t) dt \right) + w_k. \quad (78)$$

As before, from Proposition 1, all is needed to solve the MV control problem is a discrete model capable of producing  $z$  and this modified version of  $y$  as outputs.

### Proposition 3

For the DM model (16)–(18), the asynchronous measurement model (78) and the turbulence model (38)–(39), and assuming that  $A_\phi$  is a stability matrix, there exists a Gaussian white noise  $v$  independent of  $w$ , with variance  $\text{Var}(v_k) \triangleq \Sigma_v$  and matrices  $A$ ,  $B$ ,  $\Gamma$ ,  $G$ ,  $C$ ,  $\Delta$  such that

$$x_{k+1} = Ax_k + Bu_k + \Gamma v_k, \quad (79)$$

$$z_k = Gx_k, \quad (80)$$

$$y_k = Cx_{k-1} + \Delta u_{k-1} + w_k, \quad (81)$$

where

$$x_k \triangleq \begin{pmatrix} x_{\phi, k+1} \\ \overline{\phi}_{k+1}^{\text{tur}, 1} \\ \overline{\phi}_{k+1}^{\text{tur}, 2} \\ \overline{\phi}_{k+1}^{\text{tur}} \\ \overline{\phi}_k^{\text{tur}, 2} \\ x_{m, k} \\ x_{m, k-1} \\ u_{k-1} \end{pmatrix} \quad (82)$$

with

$$\overline{\phi}_{k+1}^{\text{tur},1} \triangleq \frac{1}{T_s - T_c} \int_{kT_s}^{(k+1)T_s - T_c} \phi^{\text{tur}}(t) dt, \quad (83)$$

$$\overline{\phi}_{k+1}^{\text{tur},2} \triangleq \frac{1}{T_c} \int_{(k+1)T_s - T_c}^{(k+1)T_s} \phi^{\text{tur}}(t) dt. \quad (84)$$

Expressions for  $A$ ,  $B$ ,  $\Gamma$ ,  $G$ ,  $C$ ,  $\Delta$  and  $\Sigma_v$  are given in Appendix B.

*Proof*

As in the proof of Proposition 2, the expression for  $G$  is obvious from the definition of  $\overline{\phi}^{\text{tur},1}$  and  $\overline{\phi}^{\text{tur},2}$  and identity

$$\overline{\phi}_k^{\text{tur}} = \frac{T_s - T_c}{T_s} \overline{\phi}_k^{\text{tur},1} + \frac{T_c}{T_s} \overline{\phi}_k^{\text{tur},2}. \quad (85)$$

The starting point for computing  $C$  is to note that from (78),

$$y_k = D \left( \frac{T_c}{T_s} \overline{\phi}_{k-1}^{\text{tur},2} + \frac{T_s - T_c}{T_s} \overline{\phi}_k^{\text{tur},1} \right) - D \left( \frac{1}{T_s} \int_{(k-1)T_s - T_c}^{kT_s - T_c} \phi^{\text{cor}}(t) dt \right) + w_k \quad (86)$$

$$= D \left( \frac{T_c}{T_s} \overline{\phi}_{k-1}^{\text{tur},2} + \frac{T_s - T_c}{T_s} \overline{\phi}_k^{\text{tur},1} \right) - \frac{1}{T_s} DN \left( \int_{(k-1)T_s - T_c}^{(k-1)T_s} p(t) dt + \int_{(k-1)T_s}^{kT_s - T_c} p(t) dt \right) + w_k. \quad (87)$$

Using (36), and evaluating the resulting matrix integrals, this leads to formulae for  $C$ ,  $\Delta$  in Appendix B. As for the synchronous case, the part of  $A$ ,  $B$  corresponding to discrete delays is obvious. The non-trivial remaining part of the proof is the determination of an equivalent stochastic discrete model for the quadruplet

$$\zeta_k \triangleq \begin{pmatrix} \zeta_{k,1} \\ \zeta_{k,2} \\ \zeta_{k,3} \\ \zeta_{k,4} \end{pmatrix} \triangleq \begin{pmatrix} x_{\phi,k+1} \\ \overline{\phi}_{k+1}^{\text{tur},1} \\ \overline{\phi}_{k+1}^{\text{tur},2} \\ \overline{\phi}_{k+1}^{\text{tur}} \end{pmatrix}. \quad (88)$$

A direct corollary of the proof of Proposition 2 is that for any sampling period  $T_s$ , there exists a discrete Gaussian white noise  $v$  sequence such that at all times  $t = kT_s$ ,

$$x_{\phi,k+2} = e^{T_s A_\phi} x_{\phi,k+1} + v_{1,k}, \quad (89)$$

$$\overline{\phi}_{k+2}^{\text{tur}} = M_1(T_s) x_{\phi,k+1} + v_{2,k}, \quad (90)$$

$$\overline{\phi}_{k+2}^{\text{tur}} = M_2(T_s) x_{\phi,k+1} + v_{3,k}, \quad (91)$$

where  $M_1$  and  $M_2$  are matrix-valued functions of  $T_s$  given in Appendix B. Since this property holds for any sampling period, it can also be applied between times  $t = kT_s$  and  $t = (k+1)T_s - T_c$  – that is, with a fictitious sampling period  $T_s - T_c$ . This establishes that there exists a Gaussian vector  $v_k^1$ , independent of  $x_{\phi,k+1} = x_{\phi}((k+1)T_s)$ , such that

$$x_{\phi}((k+1)T_s - T_c) = e^{(T_s - T_c)A_\phi} x_{\phi,k+1} + v_{1,k}^1, \quad (92)$$

$$\overline{\phi}_{k+2}^{\text{tur},1} = M_1(T_s - T_c) x_{\phi,k+1} + v_{2,k}^1, \quad (93)$$

$$\overline{\phi}_{k+2}^{\text{tur},1} = M_2(T_s - T_c) x_{\phi,k+1} + v_{3,k}^1, \quad (94)$$

where

$$\overline{\phi}_{k+2}^{\text{tur},1} \triangleq \frac{1}{T_s - T_c} \int_{(k+1)T_s}^{(k+2)T_s - T_c} e^{(t-kT_s)A_m^T} C_m^T N^T \phi^{\text{tur}}(t) dt. \quad (95)$$

Let us now apply the same construction procedure between times  $t = (k+1)T_s - T_c$  and  $t = (k+2)T_s$ , with pseudo-sampling period  $T_c$ . Hence there exists a Gaussian vector  $v_k^2$ , independent of  $x_\phi((k+1)T_s - T_c)$  – and also, crucially, of  $v_k^1$  – such that

$$x_{\phi,k+2} = e^{T_c A_\phi} x_\phi((k+1)T_s - T_c) + v_{1,k}^2, \quad (96)$$

$$\overline{\phi}_{k+2}^{\text{tur},2} = M_1(T_c) x_\phi((k+1)T_s - T_c) + v_{2,k}^2, \quad (97)$$

$$\overline{\phi}_{k+2}^{\text{tur},2} = M_2(T_c) x_\phi((k+1)T_s - T_c) + v_{3,k}^2, \quad (98)$$

where

$$\overline{\phi}_{k+1}^{\text{tur},2} \triangleq \frac{1}{T_c} \int_{(k+2)T_s - T_c}^{(k+2)T_s} e^{(t-(k+1)T_s + T_c)A_m^T} C_m^T N^T \phi^{\text{tur}}(t) dt. \quad (99)$$

From the definitions of  $\overline{\phi}_{k+2}^{\text{tur},1}$  and  $\overline{\phi}_{k+2}^{\text{tur},2}$ , it is immediately checked that

$$\overline{\phi}_{k+2}^{\text{tur}} = \frac{T_s - T_c}{T_s} \overline{\phi}_{k+2}^{\text{tur},1} + \frac{T_c}{T_s} e^{(T_c - T_s)A_m^T} \overline{\phi}_{k+2}^{\text{tur},2}. \quad (100)$$

Combining the two steps into a single iteration, all coordinates of  $\zeta_{k+1}$  can now be expressed as linear functions of  $\zeta_k$ ,  $v_k^1$  and  $v_k^2$ . Setting

$$v_k \triangleq \begin{pmatrix} v_k^1 \\ v_k^2 \end{pmatrix} \quad (101)$$

and noting that by construction  $v$  is a Gaussian white noise, with  $v_k$  independent of  $\zeta_k$ , this results in the expressions for  $A$ ,  $\Gamma$  in Appendix B.  $\square$

## 6. TIP-TILT MIRROR CONTROL FOR THE EUROPEAN EXTREMELY LARGE TELESCOPE

In this section, the MV control design procedure for the synchronous case is applied to atmospheric tip/tilt compensation for a DM with first-order dynamics. In AO systems, the so-called tip-tilt modes correspond to the two degrees of freedom rotation of a flat wavefront around the center of the field of view. These two modes can be corrected, together with the other turbulent modes, by a DM with deformable surface. They can also be handled separately by a dedicated tip-tilt mirror (TTM), which is essentially a mobile flat mirror (an equivalent alternative is to mount the DM on a two-degrees of freedom mobile tip-tilt platform). In practice, the use of a dedicated tip-tilt actuator enables to ‘off-load’ a part of the control effort from the DM. For a given level of turbulence energy, this enables to use DMs with lower maximum stroke—or, conversely, to compensate effectively more energetic turbulence.

Such two-stage designs have been retained for the future ELTs, for which the tip-tilt modes concentrate a very large part of the turbulent phase energy, and where AO systems are required to deliver very high levels of variance reduction. However, since TTMs necessarily involve more ponderous moving parts, they naturally tend to exhibit slower and/or more resonant dynamics than DMs with deformable surface. As a result, the corresponding optimal control problem has been recognized as a relevant challenge from an applicative point of view.

The stochastic model for tip-tilt turbulent modes used in this section is representative of operational conditions for the European ELT. The model has been fitted based on the turbulent modes’

PSD, which has been derived from physically realistic time-domain simulations provided by the European Southern Observatory. Because this physical model features a near-total decoupling between the otherwise statistically identical tip and tilt modes, and because the TTM's dynamics, in the first approximation, can also be assumed to be decoupled, the optimal control problem is treated here in a fully decoupled way, i.e. as a SISO control problem.

The model fitting was performed for a telescope pupil diameter of 42 m; the optimal parameters are an atmospheric seeing of 1.1 mas@0.5  $\mu\text{m}$ , an outer scale of 100 m and a wind speed of 10 m/s. This results for the tip or tilt modes in the scalar continuous-time first-order stochastic model

$$d\phi^{\text{tur}}(t) = a_\phi \phi^{\text{tur}}(t) dt + d\eta(t) \quad (102)$$

with  $a_\phi = -4.76 \cdot 10^{-3}$  and  $\Sigma_\eta = 7.96$ . This corresponds to (38)–(39), with

$$x_\phi = \phi^{\text{tur}}, \quad A_\phi = a_\phi, \quad C_\phi = 1. \quad (103)$$

For the DM's dynamics, we consider a first-order model with unitary DC gain and time constant  $\tau \geq 0$ :

$$p(t) + \tau \dot{p}(t) = u(t). \quad (104)$$

As this is a scalar problem, it can be assumed with no loss of generality that the DM influence matrix is unitary, so that  $\phi^{\text{cor}} = p$ . Thus, the DM model corresponds to (16)–(18), with

$$x_m = p, \quad A_m = -\frac{1}{\tau}, \quad B_m = \frac{1}{\tau}, \quad C_m = 1, \quad D_m = 0, \quad N = 1. \quad (105)$$

In this special case, the statistic  $\bar{\varphi}_{k+1}^{\text{tur}}$  is defined as

$$\bar{\varphi}_{k+1}^{\text{tur}} = \frac{1}{T_s} \int_{kT_s}^{(k+1)T_s} \exp\left(-\frac{t-kT_s}{\tau}\right) \phi^{\text{tur}}(t) dt \quad (106)$$

while the discrete model enabling to solve the equivalent discrete LQ problem has 7 coordinates.

The sampling period being set to  $T_s = 2\text{ ms}$ , MV controllers were computed and their performance assessed for DM time constant  $\tau$  ranging from 0 to 0.4 s. For each value of  $\tau$ , two MV controllers were computed: the full information controller and the incomplete information controller corresponding to a realistic signal-to-noise ratio. In addition, for each value of  $\tau$ , closed-loop optimal performance was compared with the performance of the suboptimal controller in Section 2, which in this case simply corresponds to

$$u_k = \hat{\phi}_{k+1|k}^{\text{tur}}, \quad (107)$$

where  $\hat{\phi}_{k+1|k}^{\text{tur}}$  is the prediction obtained from the Kalman filter attuned to the infinitely fast DM case. It is important to note that this Kalman filter becomes a suboptimal observer for a DM with first-order dynamics, since it does not properly take into account the effect of past controls on the actual measurement. Thus, in the incomplete information case, two sources of performance degradation operate: suboptimal control gain and suboptimal observer.

The analytical closed-loop performance based on the variance calculations in Section 4 is compared with Monte Carlo simulations. These are conducted in time-domain using SIMULINK block-diagram where an end-to-end (E2E) simulator generates realistic turbulence trajectories, so that the residual phase variance is assessed directly from (1). It should be emphasized that the E2E simulations are not based on stochastic LTI turbulence models such as (38)–(39). Instead, they rely on so-called frozen layer models, where the turbulence is considered as static screens translating above the telescope's aperture. Despite the clear mismatch between these physically realistic simulations and the first-order LTI stochastic turbulence model, these time-domain results are in fairly good agreement with model-based residual predictions, as illustrated below.

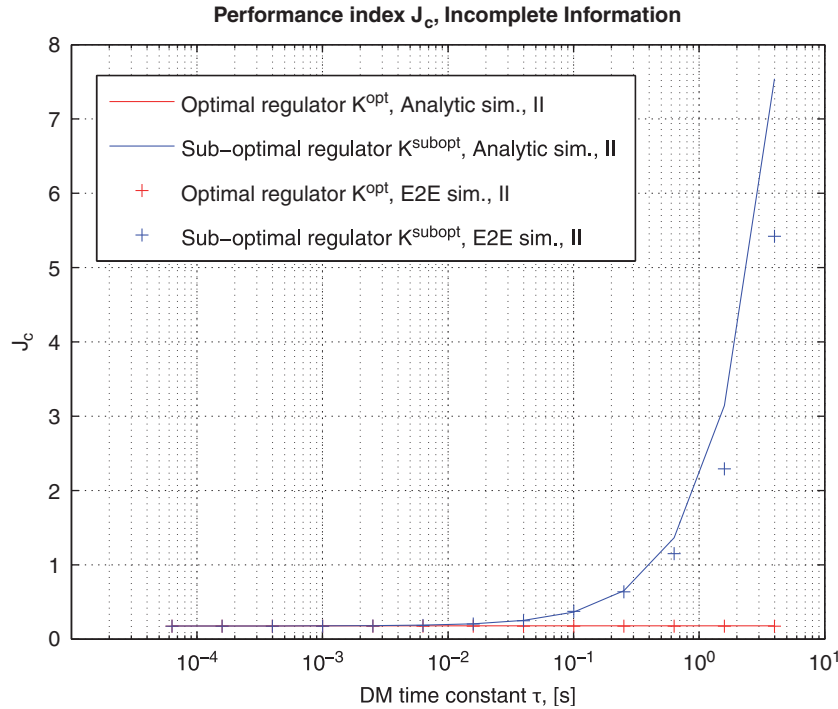


Figure 2. Residual phase variance as a function of DM's time constant  $\tau$  for the incomplete information optimal and suboptimal (infinitely fast DM) controllers. Crosses indicate performance evaluated through time-domain simulations with turbulence trajectories generated by E2E simulator. Solid lines correspond to analytical variance calculations using LTI turbulence model.

Figure 2 presents the variation of  $J^c$  as a function of the DM time constant  $\tau$  for the optimal controller and suboptimal projection in the incomplete information case. These curves show that even for large values of  $\tau$ , the MV controller maintains the same performance level as in the infinite DM case, or even better (this apparently counter-intuitive result stems from the fact that a DM with first-order dynamics controlled through a ZOH generates a richer set of trajectories than its infinitely fast counterpart). They also reveal that the performance degradation resulting from the use of the suboptimal fast DM-adapted controller remains moderate for values of  $\tau$  roughly less than  $T_s$ . As a result, this suboptimal controller may provide an acceptable alternative to the optimal but more complex MV controller for low values of  $\tau$ .

Another relevant concern is whether the optimal controller, which in a sense inverts the DM's dynamics, may generate unacceptable control efforts. This concern can be addressed by computing the average control energy, which is readily obtained from the residual variance calculations in Section 4. Figure 3 shows the evolution of the control energy for the optimal and suboptimal controllers as a function of  $\tau$  in the incomplete information case. Surprisingly, in this case, the control energy of the optimal controller only begins to shoot up when  $\tau$  is greater than 1 s, a physically unrealistic value for this mechanical device.

Finally, a portion of a typical phase, control and DM position trajectories for the optimal and suboptimal incomplete information controllers is shown in Figure 4 for a DM with  $\tau = 0.01$  s. As could be expected, the DM's trajectory generated by the optimal controller is much closer to the turbulence than for the suboptimal controller.

## 7. CONCLUSION

The results and constructions presented in this paper enable to transform the sampled data MV AO control problem, defined as the determination of the discrete controller minimizing the variance

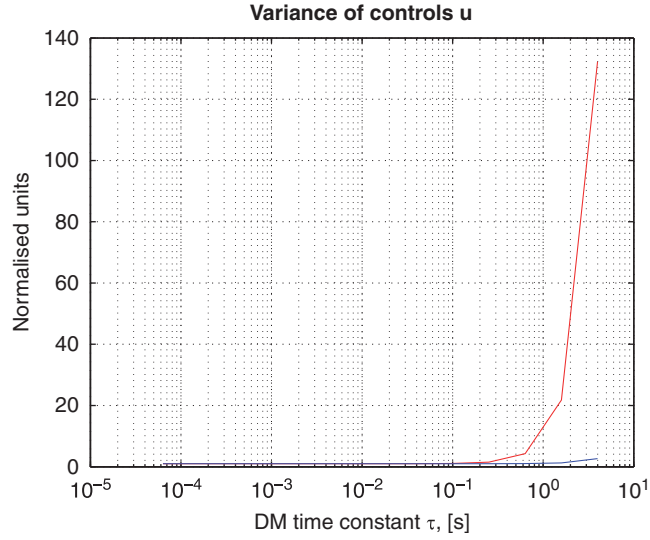


Figure 3. Average control variance as a function of DM's time constant  $\tau$  for the incomplete information optimal and suboptimal (infinitely fast DM) controllers.

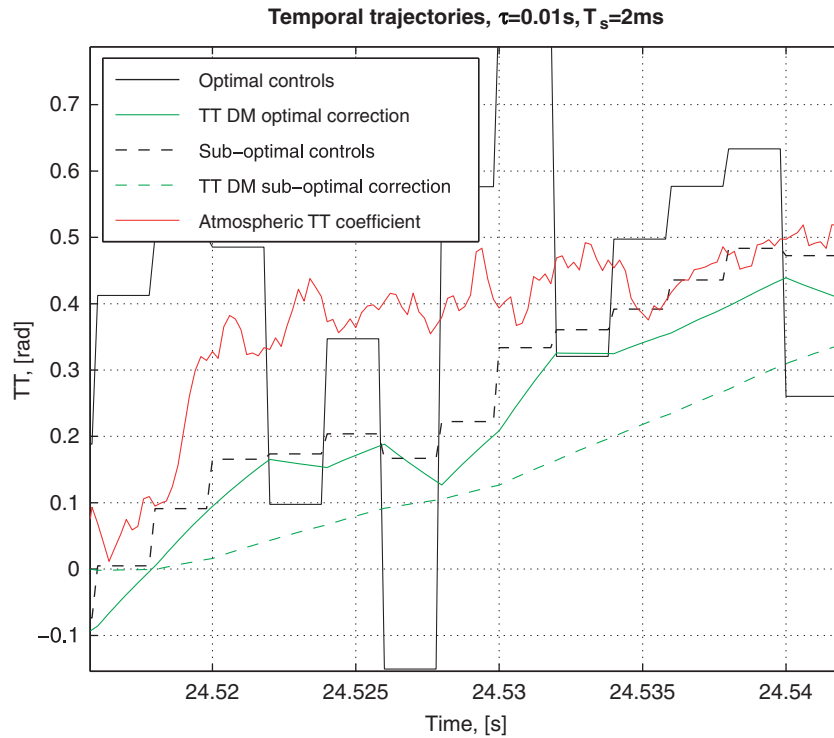


Figure 4. Details of turbulence, controls and DM's position trajectories for the incomplete information optimal and suboptimal (infinitely fast DM) controllers for  $\tau = 5T_s = 0.01$  s.

of the continuous-time residual phase, into an equivalent standard discrete-time LQG problem, which can be solved constructively using any off-the-shelf LQ/Kalman filtering software package. This approach can be applied as soon as two key ingredients are available: (a) a continuous-time state-space representation for the DM's dynamics and (b) a stochastic model for the discrete-time sufficient statistics  $(\bar{\phi}^{\text{tur}}, \bar{\varphi}^{\text{tur}})$ , which may be derived either directly from measurements and/or

available statistical priors or constructed from an underlying continuous-time stochastic model. The procedure also enables to cope with arbitrary measurement processing/control computation delays.

The critical assumption needed to achieve equivalence of the continuous- and discrete-time criterion is that the steady-state gain between the control  $u$  and the actual DM surface deformation is invertible. This covers a wide class of DM dynamics, including PDE surface deformation models. However, this restrictive assumption should be seen as minimal in the sense that, when it is relaxed, the MV control problem, in most cases of practical interest, becomes singular, with different control sequences achieving the same residual variance.

For such systems, it is therefore imperative either to change the definition of  $u$  in order to restore unicity or to add appropriate control-energy penalties to  $J^c$ . In both cases, an equivalent discrete criterion can then be easily derived. A case in point are the so-called ‘woofer-tweeter’ designs which associate two DM’s with overlapping influence functions but different combinations of stroke and bandwidth—typically, a ‘woofer’ with slow response but large stroke and a fast ‘tweeter’ with limited stroke. The MV approach enables to efficiently split the control effort between the two actuators while retaining optimal performance [22].

For more standard AO configurations, simulations conducted for simple first- and second-order DM dynamics [12, 13] suggest that the level of closed-loop performance obtained with an infinitely fast DM could in theory be maintained (or even, in the ideal case, marginally improved) even in cases where the DM’s time constants become large with respect to the sampling period. However, because the MV controller in effect seeks to invert the invertible part of the DM’s dynamics, the optimal control gain inevitably increases for slower DMs, resulting in higher control efforts.

Another important practical limitation is the often-neglected fact that ‘optimal’ control design procedures, in practice, are ultimately only as good as the underlying models upon which they rest. In the case of AO, this centrally means models capable to reproduce with sufficient accuracy the inter-sampling, and thus high-frequency, behavior of both the DM and the atmospheric turbulence. Thus, the actual design of high-performance controllers is likely to require significant research efforts in model construction, validation and parameter identification.

## APPENDIX A: HYBRID STATE-SPACE MODEL—SYNCHRONOUS CASE

The matrices for the state-space model in Proposition 2, partitioned according to the 7 components of  $x_k$ , are:

$$A \triangleq \begin{pmatrix} e^{T_s A_\phi} & 0 & 0 & 0 & 0 & 0 & 0 \\ M_1(T_s) & 0 & 0 & 0 & 0 & 0 & 0 \\ M_2(T_s) & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & I & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & e^{T_s A_m} & 0 & 0 \\ 0 & 0 & 0 & 0 & I & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 \end{pmatrix} \quad (A1)$$

with

$$\begin{pmatrix} M_1(T_s) \\ M_2(T_s) \end{pmatrix} \triangleq \begin{pmatrix} 0 & I & 0 \\ 0 & 0 & I \end{pmatrix} e^{T_s A_\zeta(T_s)} \begin{pmatrix} I \\ 0 \\ 0 \end{pmatrix}, \quad (A2)$$

where

$$A_{\zeta}(T_s) \triangleq \begin{pmatrix} A_{\phi} & 0 & 0 \\ \frac{1}{T_s} C_{\phi} & 0 & 0 \\ \frac{1}{T_s} e^{T_s A_m^T} C_m^T N^T C_{\phi} & 0 & -A_m^T \end{pmatrix} \quad (\text{A3})$$

and

$$B \triangleq \begin{pmatrix} 0 \\ 0 \\ 0 \\ 0 \\ A_m^{-1}(e^{T_s A_m} - I)B_m \\ 0 \\ I \end{pmatrix}, \quad \Gamma \triangleq \begin{pmatrix} I & 0 & 0 \\ 0 & I & 0 \\ 0 & 0 & I \\ 0 & 0 & 0 \\ 0 & 0 & 0 \\ 0 & 0 & 0 \\ 0 & 0 & 0 \end{pmatrix}, \quad (\text{A4})$$

$$G \triangleq \begin{pmatrix} 0 & I & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & I & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & I & 0 & 0 \end{pmatrix}, \quad (\text{A5})$$

$$C \triangleq (0 \quad 0 \quad 0 \quad D \quad 0 \quad -\frac{1}{T_s} D N C_m (e^{T_s A_m} - I) A_m^{-1} \dots \quad (\text{A6})$$

$$\dots D N \left( I + \frac{1}{T_s} C_m (e^{T_s A_m} - I) A_m^{-2} B_m \right)), \quad (\text{A7})$$

$$\Sigma_v \triangleq \Xi(T_s) \triangleq \int_0^{T_s} e^{t A_{\zeta}(T_s)} \begin{pmatrix} \Sigma_{\eta} & 0 & 0 \\ 0 & 0 & 0 \\ 0 & 0 & 0 \end{pmatrix} e^{t A_{\zeta}(T_s)^T} dt. \quad (\text{A8})$$

## APPENDIX B: HYBRID STATE-SPACE MODEL—ASYNCHRONOUS CASE

Let the matrix-valued functions  $M_1, M_2$  and  $\Xi$  matrices being defined as in Appendix A. Then the state-space model in Proposition 3, partitioned according to the 8 components of  $x_k$ , is defined by

$$A \triangleq \begin{pmatrix} e^{T_s A_{\phi}} & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ A_{21} & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ A_{31} & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ A_{41} & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & I & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & e^{T_s A_m} & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & I & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \end{pmatrix} \quad (\text{B1})$$

with

$$A_{21} = M_1(T_s - T_c), \quad (\text{B2})$$

$$A_{31} = M_1(T_c)e^{(T_s - T_c)A_\phi}, \quad (\text{B3})$$

$$A_{41} = \frac{T_s - T_c}{T_s} M_2(T_s - T_c) + \frac{T_c}{T_s} e^{(T_c - T_s)A_m^T} M_2(T_c)e^{(T_s - T_c)A_\phi} \quad (\text{B4})$$

and

$$B \triangleq \begin{pmatrix} 0 \\ 0 \\ 0 \\ 0 \\ 0 \\ A_m^{-1}(e^{T_s A_m} - I)B_m \\ 0 \\ I \end{pmatrix}, \quad (\text{B5})$$

$$\Gamma \triangleq \begin{pmatrix} e^{T_c A_\phi} & 0 & 0 & I & 0 & 0 \\ 0 & I & 0 & 0 & 0 & 0 \\ M_1(T_c) & 0 & 0 & 0 & I & 0 \\ \frac{T_c}{T_s} e^{(T_c - T_s)A_m^T} M_2(T_c) & 0 & \frac{T_s - T_c}{T_s} I & 0 & 0 & \frac{T_c}{T_s} e^{(T_c - T_s)A_m^T} \\ 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 \end{pmatrix}, \quad (\text{B6})$$

$$G \triangleq \begin{pmatrix} 0 & I & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & \frac{T_s - T_c}{T_s} I & \frac{T_c}{T_s} I & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & I & 0 & 0 \end{pmatrix}, \quad (\text{B7})$$

$$C \triangleq \begin{pmatrix} 0 & \frac{T_s - T_c}{T_s} D & 0 & 0 & \frac{T_c}{T_s} D & C_6 & C_7 & C_8 \end{pmatrix}, \quad (\text{B8})$$

$$\Delta \triangleq -\frac{1}{T_s} DN(T_c + C_m A_m^{-2} e^{(T_s - T_c)A_m} (e^{T_c A_m} - I)B_m), \quad (\text{B9})$$

$$\Sigma_v \triangleq \begin{pmatrix} \Xi(T_s - T_c) & 0 \\ 0 & \Xi(T_c) \end{pmatrix} \quad (\text{B10})$$

with

$$C_6 \triangleq -\frac{1}{T_s} DNC_m A_m^{-1} (e^{(T_s - T_c)A_m} - I), \quad (\text{B11})$$

$$C_7 \triangleq -\frac{1}{T_s} D N C_m A_m^{-1} e^{(T_s - T_c) A_m} (e^{T_c A_m} - I), \quad (\text{B12})$$

$$C_8 \triangleq -\frac{1}{T_s} D N (T_s - T_c + C_m A_m^{-2} (e^{(T_s - T_c) A_m} - I) B_m). \quad (\text{B13})$$

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